



Sports Data Campus

FINAL PROJECT

**Thesis Statement: “Clustering Match Behaviour to
Redefine Tactical Roles: A Data-Driven Framework for
Scouting and Player Profiling in Professional Football”**

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Abstract

This project presents a data-driven framework for tactical role profiling in professional football, developed through an unsupervised machine learning pipeline and deployed in an interactive dashboard. Drawing on data from La Liga, Ligue 1, and Serie A during the 2023–24 season, the analysis used Principal Component Analysis (PCA) and K-Means clustering to group players into interpretable tactical archetypes such as Deep-Lying Playmaker, Defensive Anchor/Stopper, Channel Runner, and Disruptive Presser.

The modular pipeline was designed for reproducibility and scalability, integrating API-Football and FBref datasets, applying per-90 normalisation, and implementing clustering at both league-specific and global levels. A Streamlit dashboard translated these outputs into accessible scouting tools, featuring role-based filters, radar charts, and player comparison modules.

Key findings show that unsupervised clustering can uncover tactically coherent roles across positions, with league-specific tendencies: Serie A produced more defensive anchors and stoppers, Ligue 1 more disruptive pressers and creative forwards, and La Liga more possession controllers. While effective, the approach faced limitations related to metric coverage, hybrid role misclassifications, and the assumptions of K-Means.

The project contributes to football analytics by operationalising role-based clustering into a scout-friendly interface, aligning with trends that prioritise functional profiling over rigid positional categories. Future work should integrate contextual and spatiotemporal data, explore alternative clustering methods, and expand analysis to team-level role profiling. Ultimately, this work demonstrates that tactical clarity and technical rigour can coexist, advancing football analytics toward greater interpretability and applied relevance.

Keywords

Football analytics; player profiling; unsupervised learning; clustering; PCA; scouting; tactical roles; dashboard; machine learning; sports data analysis; k-means clustering; principal component analysis; data visualisation; performance metrics; La Liga; Serie A; Ligue 1.

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1. Executive Summary (Step 1: Management Approach)

This project presents a tactical and data-driven approach to player role profiling in professional football, developed through an interactive application that integrates unsupervised machine learning with performance metrics. The primary objective is to construct a clustering model that identifies player archetypes based on actual match behaviour, derived from granular statistical outputs rather than traditional positional labels. The analytical pipeline results in a Streamlit-based application that enables users (including scouts, analysts and coaches) to explore player profiles across three major European leagues (La Liga, Ligue 1 and Serie A). The platform combines normalized performance data, clustering outputs and role-based labels into an accessible dashboard designed to support scouting, recruitment and tactical evaluation.

The potential impact of this project lies in its ability to make football analytics accessible by translating complex data into interpretable tactical insights. By focusing on role behaviour rather than nominal position, the tool offers a more nuanced understanding of player function within different systems and contexts. It also provides a scalable foundation for future enhancements, including the integration of contextual metrics (such as pressing intensity, spatial occupation and transitional involvement) and match-specific dynamics.

In addition to its practical utility, the project contributes to the growing body of research on unsupervised learning in football analytics, particularly in the domain of tactical role identification and player modelling. It aligns with recent academic efforts to move beyond traditional metrics and explore behavioural clustering as a means of understanding player function (Rein and Memmert, 2023).

Key outcomes include a clean and labelled dataset of tactical roles derived from unsupervised clustering, an interactive application that visualizes player behaviour across leagues and positions, a methodological pipeline aligned with real-world scouting workflows and a portfolio-ready product that reflects both technical proficiency and footballing insight.

This project reflects my commitment to bridging the gap between data science and football strategy and serves as a foundation for future work in tactical modelling, scouting support and football analytics innovation. The outcomes are directly applicable to modern scouting and recruitment workflows, where data-driven profiling increasingly complements traditional video scouting.

2. Introduction (Step 2: Macro Vision)

The increasing integration of data analytics into professional football has transformed how clubs approach recruitment, tactical planning and performance evaluation. Yet, despite the proliferation of data, many analytical tools remain inaccessible to practitioners without technical backgrounds. This project responds to that challenge by developing an interactive application that translates raw performance data into interpretable tactical profiles, enabling decision-makers to assess players based on how they behave on the pitch rather than by nominal position.

My work is situated within the broader context of football's evolving analytical landscape, where clustering methodologies and role-based profiling are increasingly used to uncover patterns and inform strategic decisions. By focusing on unsupervised learning and tactical interpretability, I aim to contribute a tool that not only performs technically but also aligns with the practical realities of scouting and coaching.

The project focuses on the 2023 to 2024 season and includes players from three major European leagues (La Liga in Spain, Ligue 1 in France and Serie A in Italy). These competitions were selected for their tactical diversity and stylistic contrasts, which support the goal of identifying role profiles across varied footballing contexts. For each league, the analysis begins with midfielders, followed by defenders and forwards. To ensure statistical robustness, only players with a minimum of 540 minutes played (equivalent to six full matches) were retained.

Data was sourced through a hybrid approach. Initially, I considered scraping player statistics from FBref using Python's `pandas.read_html` method. However, due to FBref's transition to JavaScript-rendered content, this method proved unreliable. To ensure scalability and consistency, I transitioned to structured data access via the API-Football platform. A paid subscription enabled full access to player-level statistics across multiple leagues and seasons. To supplement the API's limitations (particularly in metrics such as pass accuracy by range), I manually exported additional data from FBref. This hybrid strategy resulted in a more complete and tactically relevant dataset.

Each dataset was filtered and processed to extract key performance indicators, including duels, recoveries, progressive passes, assists, dribbles and defensive actions. Metrics were normalized per 90 minutes and scaled prior to clustering. Players were grouped using algorithms such as K-Means, and each cluster was labelled based on its statistical signature (for example, box-to-box midfielder, ball-winning six or deep distributor). These labels were informed by both quantitative outputs and qualitative footballing knowledge.

My thesis argues that ***“Unsupervised machine learning, when applied to normalized performance metrics, can produce interpretable tactical role profiles that enhance scouting workflows and support data-driven decision-making in professional football.”*** This position aligns with Rein & Memmert (2023), who emphasise that role-based modelling provides more actionable insights than positional categorisation, and complements work by Decroos et al. (2020), who showed the predictive power of contextual player metrics.

In addition to its practical relevance, the project contributes to the academic discourse on player modelling and tactical role identification. It builds on recent studies that advocate for data-driven approaches to understanding player behaviour and tactical fit (Liu et al., 2022; Decroos et al., 2020). By prioritizing tactical relevance and accessibility, the project aims to support more informed decisions across the football ecosystem and contribute to the ongoing evolution of football analytics.

However, despite advances in clustering methods, little research has operationalised these models into user-friendly scouting dashboards, leaving a gap between academic models and practitioner adoption, a gap this project addresses.

Personal Motivation and Professional Goals

This project also reflects my personal motivation to bridge the gap between physical performance and tactical understanding. With a background in sports therapy and over a decade of experience in healthcare, I bring a perspective that connects physiological demands with data interpretation. Throughout this Master's program, I have developed core competencies in Python, data analysis and football-specific metrics. This thesis represents the culmination of those skills, applied in a real-world analytical pipeline that mirrors modern scouting and tactical evaluation.

In parallel with my academic development, I currently work as a video scout, a role that has deepened my understanding of tactical systems, player behaviour and match dynamics. My responsibilities include tagging player actions, identifying role-specific patterns, conducting opposition analysis and preparing match reports for technical staff. These tasks have sharpened my ability to interpret performance data in context and have directly influenced the way I structured and labelled player profiles within this project.

Beyond its technical scope, the application developed in this thesis serves as a portfolio piece that showcases both my analytical capabilities and footballing insight. It is also a strategic step toward transitioning into a remote data analyst or scouting-focused role within the football industry. My goal is to contribute to the development of tools and methodologies that support objective, role-based decision-making, and to help clubs and analysts make sense of complex data through interpretable tactical frameworks.

3. Objectives

The primary objective of this project is to develop a football analytics application that enables tactical role profiling through unsupervised learning and performance data. The aim is to move beyond traditional positional labels and instead classify players based on their actual match behaviour, using clustering techniques that reflect tactical function rather than nominal role.

This project is guided by the following research question: can unsupervised clustering of performance metrics reveal tactical player roles that are more informative and contextually relevant than conventional positional categories?

To address this question, I established a set of specific objectives:

First, to collect and preprocess player-level performance data from multiple sources (including API-Football and FBref), ensuring coverage across key tactical metrics such as duels, recoveries, progressive passes, assists, dribbles and defensive actions. This includes normalizing all features per 90 minutes and scaling them to allow for cross-positional comparability.

Second, to implement unsupervised machine learning algorithms (including Principal Component Analysis and K-Means clustering) that group players based on statistical similarity. These clusters are then interpreted and labeled using domain knowledge, resulting in role profiles such as box-to-box midfielder, ball-winning six or deep distributor.

Third, to design and deploy an interactive application using Streamlit that allows users to explore player profiles across three major European leagues (La Liga, Ligue 1 and Serie A). The app includes filtering tools, radar charts and similarity engines that support scouting, recruitment and tactical evaluation.

Fourth, to ensure that the outputs of the model are interpretable and accessible to both technical and non-technical stakeholders. This involves careful selection of features, clear role labelling and intuitive visualizations that align with the practical needs of analysts, coaches and scouts.

Fifth, to evaluate the model's success based on interpretability, consistency across leagues and alignment with known tactical roles. This includes assessing whether the clusters reflect meaningful footballing archetypes and whether they can be applied across different tactical systems.

Finally, to produce a clean and labelled dataset of tactical roles that can be used for further research, scouting support or football app development. This dataset serves as the foundation for future enhancements, including the integration of contextual metrics, match-specific dynamics and advanced modules such as pose detection or set-piece analysis.

In addition to these objectives, success will be evaluated on interpretability (role labels understood by practitioners), tactical validity (alignment with football logic), and scalability (cross-league applicability).

These objectives reflect my commitment to bridging the gap between data science and football strategy, and to contributing a tool that is both methodologically rigorous and tactically meaningful.

4. Conceptual and Technological Architecture

The infrastructure of this project is built upon a robust, end-to-end data pipeline designed to transform raw football statistics into interpretable tactical insights. The architecture is organized into three core stages: data acquisition, data processing and modelling, and the application layer. This tiered structure ensures scalability, transparency, and alignment with real-world scouting workflows.

4.1 Data Acquisition

The foundation of the project relies on structured, high-quality data that reflects player behaviour across multiple competitions. Rather than relying on manual scraping from web pages such as FBref (which is prone to disruption due to frequent site changes), the project adopted a more stable and professional approach.

API-Football served as the primary data source. Through programmatic access to its endpoints, player-level statistics were retrieved across three major European leagues (La Liga, Ligue 1, and Serie A). This method ensured consistency, reproducibility, and scalability across seasons and competitions.

To address limitations in the API (particularly the absence of granular passing metrics and progressive actions), manually exported data from FBref was integrated as a supplementary source. This hybrid strategy resulted in a more complete and tactically relevant dataset, suitable for clustering and role profiling.

Data Acquisition Pipeline

API-Football (automated via Python) → FBref (manual export for supplementary metrics) → Combined Dataset → Preprocessing and Feature Selection (filtering by minutes played ≥ 540 , per-90 normalization, tactical metric selection) → Modelling Pipeline (PCA, K-Means clustering, role labelling) → Streamlit Application (interactive filters, radar charts, player comparison engine).

Overview of the data acquisition pipeline, showing the integration of API-Football and FBref sources into a unified dataset for tactical modelling. The data is filtered, normalized, and processed before being clustered and visualized in the Streamlit application.

4.2 Data Processing and Modelling

This stage represents the analytical core of the project, where raw data is cleaned, transformed, and modelled to uncover tactical patterns and player archetypes.

The entire pipeline was developed in Python, selected for its flexibility, extensive library support, and suitability for football analytics workflows.

Pandas was used for data manipulation, including filtering players by minutes played (minimum 540), calculating per-90 metrics, and handling missing values. This ensured statistical robustness and comparability across roles.

NumPy supported numerical operations, including percentage calculations and feature scaling, helping prevent errors such as division by zero.

Scikit-learn was central to the machine learning phase. It enabled dimensionality reduction through Principal Component Analysis (PCA) and clustering via K-Means, which grouped players based on statistical similarity. The number of clusters was determined through iterative testing using the Elbow Method and Silhouette Score, followed by domain-informed interpretation.

Each cluster was labelled using a combination of statistical signature and tactical domain knowledge, resulting in role profiles such as box-to-box midfielder, deep distributor, or ball-winning six. These labels reflect actual match behaviour rather than nominal position.

As shown in Figure 1 below, the PCA plot illustrates the distribution of players across the first two principal components, revealing distinct clusters based on performance metrics. Each point represents an individual midfielder, color-coded by cluster assignment. The visual separation between groups supports the tactical role profiling described in Section 5.

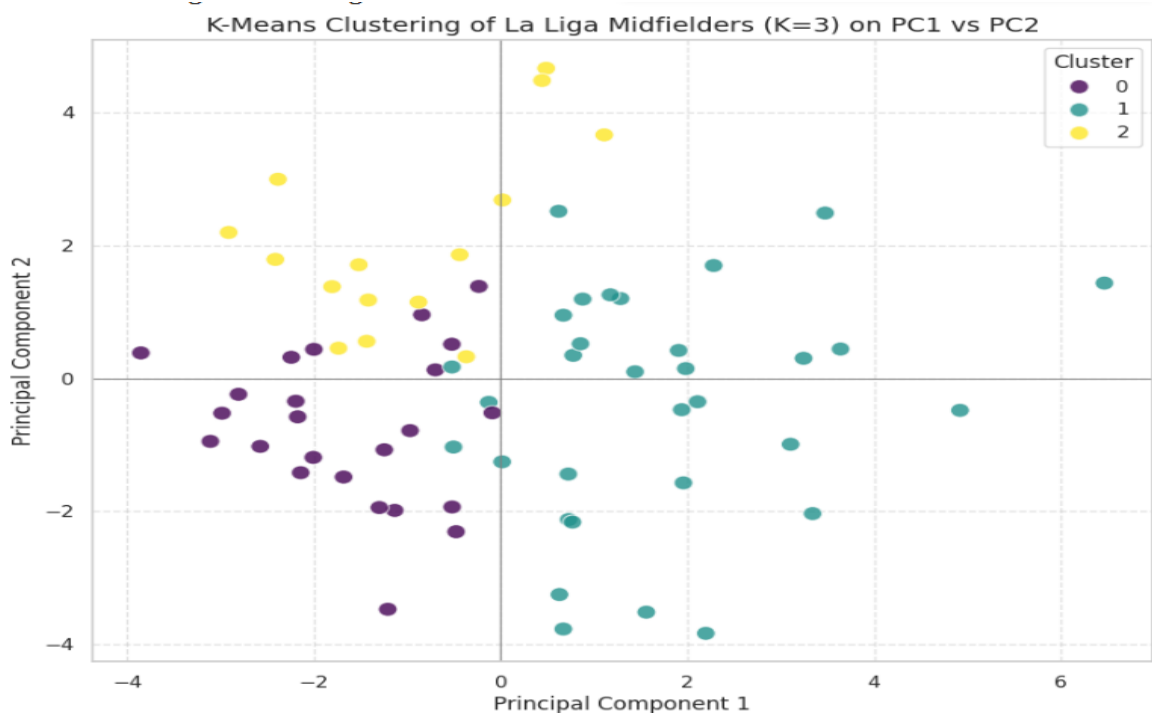


Figure 1: PCA visualization of La Liga midfielders clustered using K-Means (K=3). Each point represents a player plotted along Principal Component 1 and Principal Component 2. Color-coded clusters reflect tactical role groupings based on performance metrics.

The pipeline was designed to be modular and reproducible. Each step (from data acquisition to clustering) was documented and executed within Jupyter Notebooks, ensuring transparency and facilitating future replication or adaptation.

The analytical pipeline was developed and tested within Google Colab, which provided a flexible, cloud-based environment for data exploration, preprocessing and modelling. The final application interface was built using Streamlit and deployed via Visual Studio Code, allowing for local development and modular integration of clustering outputs and player profiles. Streamlit was selected over alternatives such as Dash and Flask because of its rapid deployment, intuitive design, and lower entry barrier for scouts and coaches without technical backgrounds.

4.3 Application Layer

The final layer of the architecture focuses on accessibility, tactical usability, and modular design. Insights generated from the modelling phase are delivered through an interactive web-based dashboard, developed using Streamlit. This framework was selected over alternatives such as Dash or Flask due to its rapid deployment capabilities, minimal setup requirements, and suitability for football-specific data exploration.

The dashboard supports positional modularity, allowing users to select between Defenders, Midfielders, and Forwards. Each module is tailored to the tactical and statistical demands of its role group. Filters enable users to select tactical roles, isolate individual players, and choose specific metrics for comparison. A dropdown menu allows side-by-side player comparisons via radar charts and bar charts, visualizing percentile ranks across selected performance dimensions.

Users can toggle between raw metrics and PCA-derived Individual Key Performance Areas. For defenders, an additional option to view grouped Key Performance Areas simplifies interpretation across a broader metric set. A dedicated metric selector provides granular control over which metrics are visualized, supporting both exploratory analysis and targeted scouting workflows.

To enhance interpretability, an expandable section provides detailed explanations of each tactical role and performance area. The interface was designed with end-users in mind—scouts, analysts, and coaches—prioritizing clarity, tactical relevance, and ease of navigation. The following screenshots illustrate the dashboard's interactive components, including role-based filtering, cluster assignments, and performance metric access across leagues and positions.

Player Dashboard

Select Dashboard

Defenders

Midfielders

Forwards

Defenders Dashboard Filters

Select Role(s)

Positional D... x

Ball-Winnin... x

Defensive A... x

Ball-Playing ... x

Compare Players (Radar + Bar)

Choose options



Tactical Dashboard for Defenders

>  Tactical Role & Key Performance Areas Explanations



Player Table

	Player Name	League	Club	Position	Tactical Role	Goals per 90	Shots	Ass
0	F. Fazio	Serie A	Salernitana	Defender	Positional Defender	0.07	6	
1	Mário Rui	Serie A	Napoli	Defender	Ball-Winning Full-Back	0	5	
2	D. D'Ambrosio	Serie A	Monza	Defender	Defensive Anchor / Stopper	0	10	
3	Pedro Pereira	Serie A	Monza	Defender	Ball-Winning Full-Back	0	1	
4	A. Florenzi	Serie A	AC Milan	Defender	Ball-Winning Full-Back	0.05	26	
5	D. Calabria	Serie A	AC Milan	Defender	Defensive Anchor / Stopper	0.04	16	
6	R. Tolói	Serie A	Atalanta	Defender	Ball-Winning Full-Back	0	3	
7	P. Hatzidiakos	Serie A	Cagliari	Defender	Positional Defender	0	3	
8	F. Ebosele	Serie A	Udinese	Defender	Ball-Winning Full-Back	0	12	

Figure 2: Dashboard interface for defenders, showing cluster-based tactical role assignments. Users can filter by role archetype (e.g., Positional Defender, Ball-Winning Full-Back) and view key performance metrics such as goals per 90, shots, and assists. Position dashboards, Tactical roles or individual players can be filtered and selected from the dropdowns on the left.

Compare Players (Radar + Bar)

Choose options

A. Cambiaso

A. Dossena

A. Florenzi

A. Gallo

A. Hakimi

A. Masina

A. Murillo

A. Obert

Average Metric Ranks by Role (0-100)

What these numbers mean: This table shows the average percentile rank for each metric within a specific role. A percentile rank is a value from 0 to 100, indicating how a player performs relative to all other players in the dataset.

Example: A value of 29.35 for Goals per 90 means the average player in this role performs better than only 29.35% of all players in that metric. This is expected for roles like a Defender, where scoring is not a primary function. A high number, such as 95.00 for a metric like Tackles Success %, would show that the average player in this role is among the best in the dataset for that skill.

Tactical Role	Goals per 90	Shots	Assists per 90	Dribble Success %	Total Pass Accuracy %	Tac
Ball-Playing Centre-Back	29.35	28.85	7.87	70.39	34.79	
Ball-Winning Full-Back	35.05	35.25	48.32	40.84	70.81	

Figure 3: Average Metric Ranks visualization for clustered players. This chart displays the relative ranking of each cluster across selected performance metrics, allowing users to compare tactical strengths and weaknesses. Metrics are normalized and ranked to enhance interpretability across roles. Position dashboards, Tactical roles or individual players can be filtered and selected from the dropdowns on the left

Combined Radar Chart (Percentile)

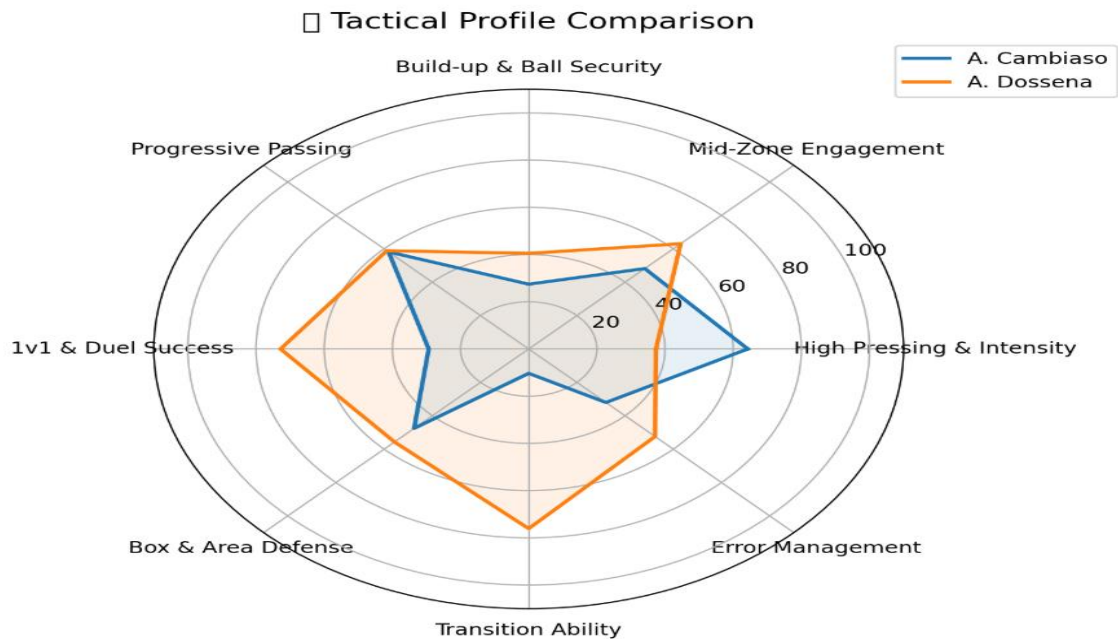


Figure 4: Radar chart displaying the average performance profiles of clustered players. Each polygon represents a tactical role archetype derived from K-Means clustering, plotted across normalized metrics such as passes, duels, interceptions, and dribbles. The chart enables visual comparison of role-specific strengths and stylistic tendencies. The players for comparison can be selected from the dropdown on left, as well as the type of the metrics (Key Performance Areas Groups, Individual Key Performance Areas, Raw Unscaled Metrics).

Player Metric Comparison

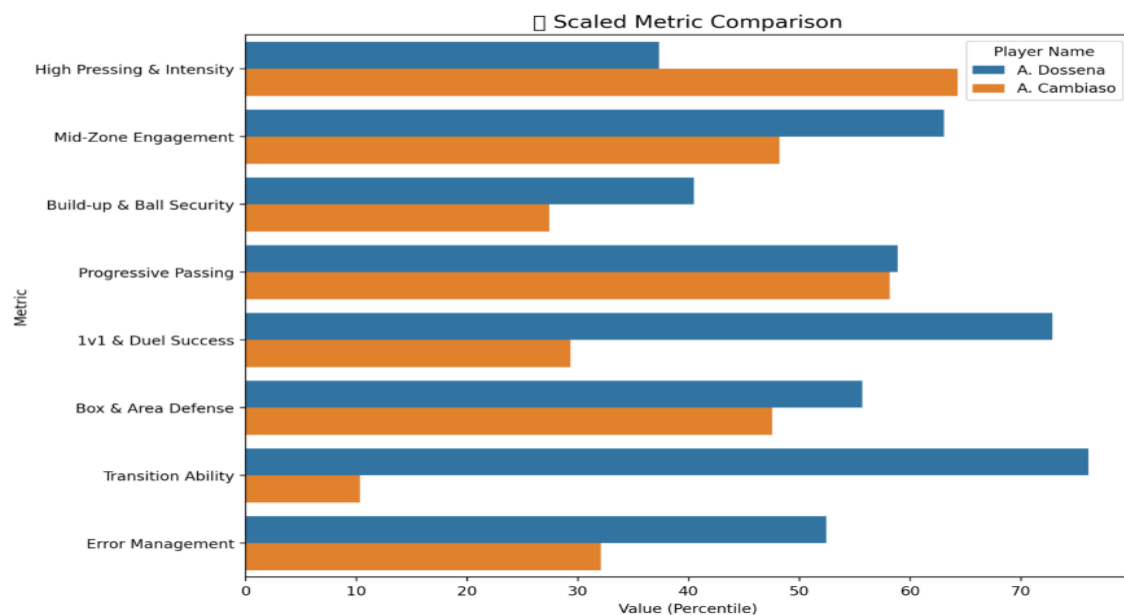


Figure 5: Bar chart comparing average values of selected performance metrics across clustered players. Each bar represents a cluster's mean score for a given metric, enabling direct comparison of tactical tendencies such as passing volume, defensive actions, and attacking output. The players for comparison can be selected from the dropdown on left, as well as the type of the metrics (Key Performance Areas Groups, Individual Key Performance Areas, Raw Unscaled Metrics).

Similar Player Finder

Choose Player to Find Similar Profiles

A. Cambiaso

Showing top 5 players most similar to A. Cambiaso based on selected metrics.

	Player Name	Club	League	Similarity
79	V. Le Goff	Lorient	Ligue 1	0.99
133	A. Hakimi	Paris Saint Germain	Ligue 1	0.98
43	F. Parisi	Fiorentina	Serie A	0.97
158	G. Kalulu	Lorient	Ligue 1	0.96
15	G. Ferrari	Sassuolo	Serie A	0.96

Figure 6: Player comparison interface displaying individual performance metrics relative to cluster averages. The selected midfielder is benchmarked across key tactical dimensions such as passing, duels, and defensive actions, enabling role validation and scouting-oriented evaluation.

4.4 Scalability, User Interface, and Future Enhancements

The architecture is designed to be modular and scalable, supporting both current functionality and future expansion. The interactive dashboard, built using Streamlit, enables users to filter players, view cluster-based tactical role assignments, and compare performance metrics across archetypes. This interface supports scouting workflows and tactical analysis through intuitive, role-driven navigation.

To enhance interpretability, the dashboard includes multiple visualization modules such as radar charts, bar plots, average metric ranks, and player comparison tools. These components translate complex clustering outputs into actionable insights for both technical and non-technical stakeholders.

Future enhancements may include the integration of spatial tracking data, dynamic role transitions, and contextual match events such as pressing intensity or set-piece involvement. These additions would further refine role profiling and support more granular tactical analysis. Expanding the app to include interactive match timelines or video tagging modules could also bridge the gap between data and live scouting workflows.

This architecture mirrors the workflows of professional recruitment departments, ensuring that the system is not only technically scalable but also operationally relevant to real scouting environments.

5. Methodologies and Techniques Employed

This project follows the **CRISP-DM (Cross-Industry Standard Process for Data Mining)** framework to structure the analytical workflow. CRISP-DM provides a robust, iterative methodology encompassing six phases: *Business Understanding*, *Data Understanding*, *Data Preparation*, *Modelling*, *Evaluation*, and *Deployment*. Its modularity and tactical flexibility make it particularly suitable for football analytics, where interpretability and role clarity are essential. This section uses the Global Midfielders notebook as a representative example, but the methodology was applied across multiple datasets with varying PCA setups and cluster counts

5.1 Business Understanding

The primary objective was to develop a system capable of identifying and profiling tactical roles of football players based on performance data. The focus was on players across top European leagues, with clustering used to group them into tactically meaningful archetypes. The goal was to support scouting, recruitment, and opposition analysis through interpretable role labelling.

5.2 Data Understanding

Data Sources and Raw Inputs

- **API-Football**: Provided match-level statistics such as goals, assists, duels, and passes.
- **FBref (via StatsBomb)**: Offered advanced metrics including pressures, progressive passes, and defensive actions.

These sources were selected for their coverage, consistency, and tactical depth.

✓ La Liga Midfielders API stats (filtered for >=540 mins) saved to: laliga_midfielders_2023_24_api_stats.csv

First 5 rows of the exported Midfielders API stats:

	name	team	position	minutes	duels_total	duels_won	goals	shots_total	shots_on_target	assists	dribbles_attempts	dribbles_success
140	T. Kroos	Real Madrid	Midfielder	2129.0	168.0	104.0	1.0	25.0	0	8.0	8.0	5.0
143	É. Capoue	Villarreal	Midfielder	1443.0	208.0	117.0	1.0	10.0	0	1.0	42.0	22.0
186	Rodri	Real Betis	Midfielder	1038.0	140.0	70.0	0.0	10.0	0	0.0	48.0	24.0
202	Jonathan Viera	Almeria	Midfielder	923.0	91.0	47.0	0.0	9.0	0	3.0	19.0	9.0
220	I. Rakitić	Sevilla	Midfielder	1426.0	120.0	69.0	2.0	21.0	0	1.0	17.0	8.0

Shape of the exported Midfielders API stats: (129, 20)

Figure 7: Raw Data Preview from La_Liga_Intro_midfielders.ipynb (df.head())

Manual Preprocessing and Filtering

- Unified player IDs across sources.
- Filtered players with fewer than 540 minutes played to remove noise from low-sample profiles.
- Standardized metric names and positional labels for consistency across notebooks.

```

import pandas as pd

# Load the API-exported midfielders stats.
# This file is expected to already contain only midfielders with >= 540 minutes.
try:
    df_api_midfielders = pd.read_csv("laliga_midfielders_2023_24_api_stats.csv")
    print("✅ La Liga Midfielders API stats loaded successfully.")
except FileNotFoundError:
    print("❌ Error: 'laliga_midfielders_2023_24_api_stats.csv' not found.")
    print("Please ensure this file has been generated by the 'La Liga Midfielders API Data Export' step.")
    exit()
except Exception as e:
    print(f"❌ An unexpected error occurred during file loading: {e}")
    exit()

# Filter for 'Midfielder' position and players with 540 minutes or more.
# Although the input file should already be filtered, this step re-confirms it.
midfielders_filtered = df_api_midfielders[
    (df_api_midfielders['position'] == 'Midfielder') &
    (df_api_midfielders['minutes'] >= 540)
].copy()

# Showing how many players were found
print(f"\n✅ Midfielders with 540 minutes or more found (re-confirmed): {len(midfielders_filtered)}")

# Printing the available columns in the filtered dataset
print("\nAvailable columns for filtered midfielders:")
print(midfielders_filtered.columns.tolist())

# Save the filtered dataset to a new CSV file
output_filename = "laliga_midfielders_2023_24_filtered.csv"
midfielders_filtered.to_csv(output_filename, index=False)
print(f"\n✅ Filtered midfielders dataset saved as: {output_filename}")

# Display the first few rows of the filtered dataset
print("\nFirst 5 rows of the filtered midfielders dataset:")
print(midfielders_filtered.head().to_string())

```

Figure 8: Filtering Players with ≥ 540 Minutes Played (query logic and output)

5.3 Data Preparation

Feature Selection and Normalization

Feature selection focused on metrics that reflect tactical behaviour such as goals, assists, key passes, duels, tackles, and dribbles. These were chosen to capture both attacking and defensive contributions.

Normalization was performed using `StandardScaler` rather than `MinMaxScaler`. This choice preserved the distribution shape of each metric, which is critical when applying PCA, as it relies on variance structure to identify meaningful components.

```

import pandas as pd
from sklearn.preprocessing import StandardScaler

# Load the dataset containing the preprocessed data with one-hot encoded team columns.
# This is the input for the scaling process.
try:
    df_features_to_scale = pd.read_csv("laliga_midfielders_2023_24_preprocessed_encoded_team.csv")
    print("✅ Preprocessed midfielders dataset (with encoded teams) loaded successfully for scaling.")
except FileNotFoundError:
    print("❌ Error: 'laliga_midfielders_2023_24_preprocessed_encoded_team.csv' not found.")
    print("Please ensure this file has been generated by the 'Feature Engineering: One-Hot Encoding Team Column for Midfielders' step.")
    exit()
except Exception as e:
    print(f"❌ An unexpected error occurred during file loading: {e}")
    exit()

# Identify numerical columns to be scaled.
# This should include all per-90 metrics, percentage metrics, and one-hot encoded team columns.
# Exclude identifier columns like 'name', 'position', 'name_clean', 'team_clean', and 'minutes'
# as they are not features for clustering and should not be scaled.
columns_to_exclude_from_scaling = [
    'name', 'position', 'minutes', 'name_clean', 'team_clean',
    # Original API total columns are also excluded as we use their _p90 versions or percentages
    'duels_total_api', 'duels_won_api', 'goals_total_api', 'shots_total_api',
    'shots_on_target_api', 'assists_total_api', 'dribbles_attempts_api',
    'dribbles_success_api', 'touches_total_api', 'passes_key_api', 'passes_total_api',
    'tackles_total_api', 'interceptions_total_api'
]

# Filter out columns that don't exist in the DataFrame to prevent errors
existing_cols_to_exclude = [col for col in columns_to_exclude_from_scaling if col in df_features_to_scale.columns]

# Create a DataFrame containing only the features that need to be scaled.
# This will include all the per-90 metrics, percentage metrics, and one-hot encoded team columns.
df_numerical_features = df_features_to_scale.drop(columns=existing_cols_to_exclude, errors='ignore').copy()

# Initialize the StandardScaler.
# StandardScaler will transform the data such that each feature has a mean of 0 and a standard deviation of 1.
# This is crucial for distance-based clustering algorithms like K-Means.
scaler = StandardScaler()

# Fit the scaler to the data and transform it.
# The .fit_transform() method calculates the mean and standard deviation for each feature
# and then applies the transformation. The output is a NumPy array.
scaled_features_array = scaler.fit_transform(df_numerical_features)

# Convert the scaled NumPy array back into a Pandas DataFrame.
# It's important to retain the original column names for interpretability.
df_scaled_features = pd.DataFrame(scaled_features_array, columns=df_numerical_features.columns)
print("✅ Numerical features successfully scaled using StandardScaler.")

# Save the scaled features to a new CSV file.
# This file will be the direct input for the clustering algorithms.
output_filename = "laliga_midfielders_2023_24_scaled_metrics.csv"
df_scaled_features.to_csv(output_filename, index=False)
print(f"✅ Scaled features dataset saved as: {output_filename}")

# Display summary information about the scaled dataset.
print("\n--- Summary of Scaled Features for Midfielders ---")
print(f"Scaled dataset shape: {df_scaled_features.shape}")
print(f"\nColumn types (should all be float64):")
print(df_scaled_features.dtypes.value_counts())
print(f"\nFirst 5 rows of the scaled features (values should be centered around 0):")
print(df_scaled_features.head().to_string())

```

Figures 9 & 10: Feature Selection and Scaling Pipeline (code block)

5.4 Modelling

To illustrate the modelling process, this section uses the **Global Midfielders notebook** as a representative example. However, the clustering methodology was applied across multiple datasets, each with its own PCA configuration, optimal number of clusters (K), and tactical role definitions. For instance:

- **La Liga Midfielders:** K=3, with PCA focused on possession and attacking metrics
- **Global Defenders:** K=4, with PCA emphasizing duels, clearances, and passing ability
- **Global Forwards:** K=4, focused on duels, goals and dribbles.

The modelling pipeline remained consistent across notebooks, with adaptations based on silhouette scores, tactical clarity, and visual separation.

Dimensionality Reduction with PCA

Principal Component Analysis (PCA) was applied to reduce dimensionality and enhance cluster separation. Each notebook had its own PCA setup, tailored to the available metrics and tactical focus.

In the Global Midfielders example, components were retained only if they aligned with interpretable footballing themes:

- **PC1:** Defensive involvement and ball recovery
- **PC3:** Defensive backbone and work rate
- **PC4–PC8:** Captured pressing intensity, creativity, ball-carrying, and technical quality
- **PC2** was excluded due to weak tactical interpretability, it primarily reflected match participation rather than playing style

This selection ensured that clustering was based on stylistic signals rather than availability.

```

import pandas as pd
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
import matplotlib.pyplot as plt
import seaborn as sns

# --- Load the PCA-transformed global midfielder dataset ---
try:
    df_pca = pd.read_csv("global_midfielders_2023_24_pca_transformed.csv")
    print("✅ Global PCA-transformed data loaded successfully for Silhouette Score calculation.")
except FileNotFoundError:
    print("❌ Error: 'global_midfielders_2023_24_pca_transformed.csv' not found.")
    exit()
except Exception as e:
    print(f"❌ Unexpected error during file loading: {e}")
    exit()

# --- Select principal components for clustering (PC1-PC6 based on variance threshold) ---
pca_components = [f"PC{i}" for i in range(1, 7)]
X_for_clustering = df_pca[pca_components].copy()
print(f"✅ Using {len(pca_components)} principal components (PC1-PC6) for Silhouette Score evaluation.")

# --- Calculate Silhouette Scores across K values ---
silhouette_scores = []
K_range = range(2, 10)
print(f"Calculating Silhouette Scores for K from {min(K_range)} to {max(K_range)}...")

for k in K_range:
    kmeans = KMeans(n_clusters=k, init='k-means++', random_state=42, n_init=10)
    labels = kmeans.fit_predict(X_for_clustering)
    score = silhouette_score(X_for_clustering, labels)
    silhouette_scores.append(score)

print("✅ Silhouette Scores computed.")

```

Figure 11: PCA Explained Variance Plot (code block)

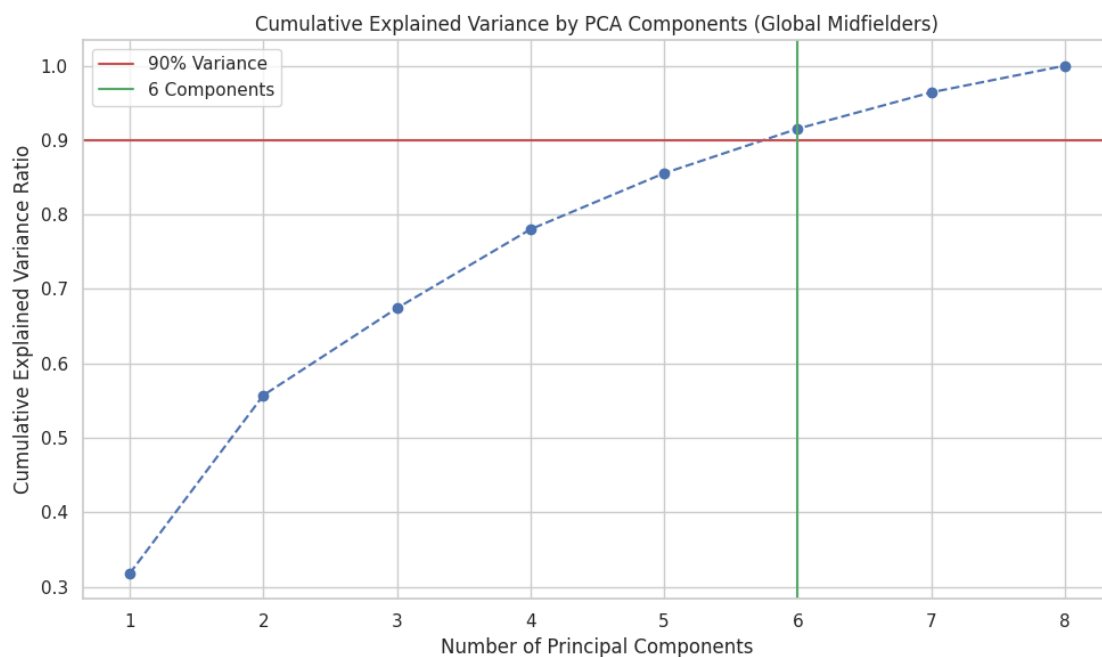


Figure 12: PCA Explained Variance Plot

Clustering with K-Means

Once dimensionality was reduced, clustering was performed on the PCA-transformed data to identify tactical groupings.

K-Means clustering was applied to the PCA-reduced data. The optimal number of clusters varied across notebooks:

- **La Liga Midfielders:** K=3
- **Global Defenders:** K=4
- **Global Forwards:** K=4 selected for clarity, though K=3 was also tested

For Global Midfielders, K=3 was chosen due to:

- A clear elbow point in the inertia plot
- Strong silhouette cohesion
- Reduced overlap compared to K=4
- Tactical readability across three distinct roles

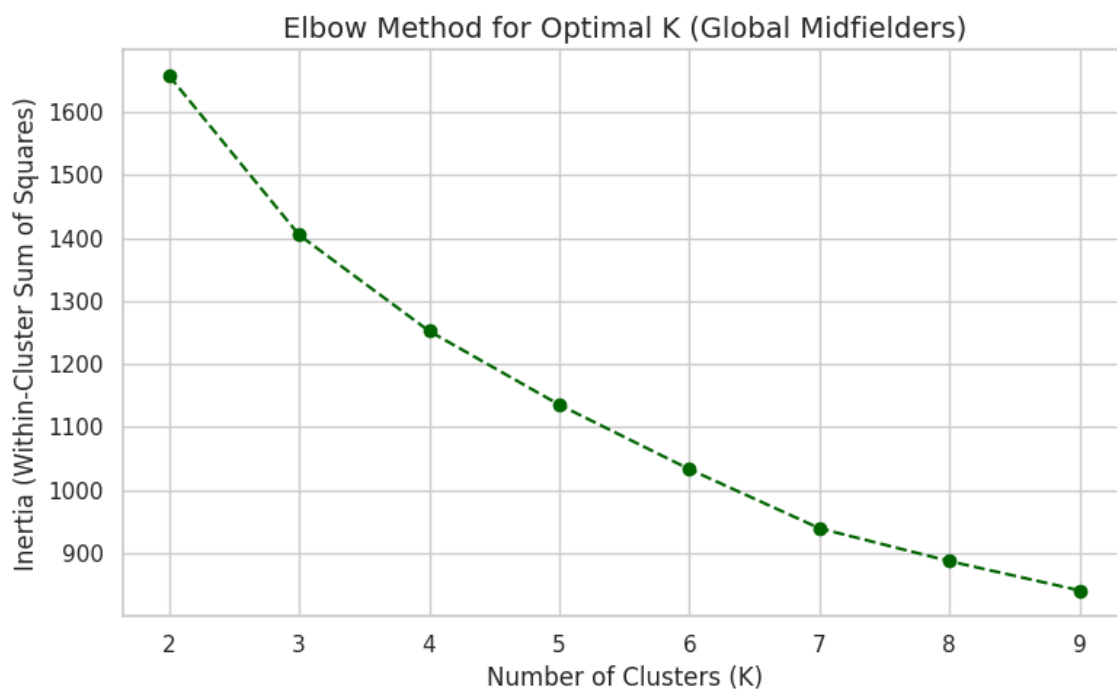


Figure 13: Elbow Method Plot for Optimal K

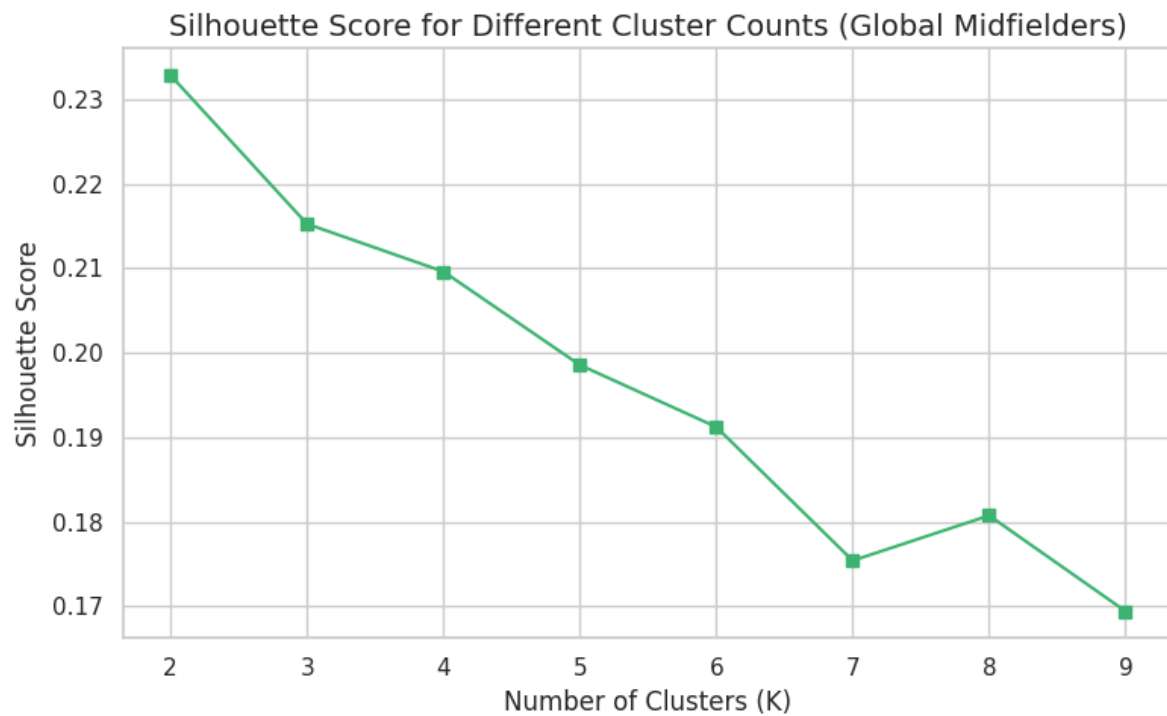


Figure 14: Silhouette Score Plot

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

# --- Load PCA-transformed global midfielder data ---
try:
    df_pca = pd.read_csv("global_midfielders_2023_24_pca_transformed.csv")
    print("✅ PCA-transformed midfielder data loaded.")
except FileNotFoundError:
    print("❌ File 'global_midfielders_2023_24_pca_transformed.csv' not found.")
    exit()
except Exception as e:
    print(f"❌ Unexpected error: {e}")
    exit()

# --- Load scaled tactical metrics ---
try:
    df_scaled_features = pd.read_csv("global_midfielders_2023_24_scaled_metrics_final.csv")
    print("✅ Scaled metrics loaded for centroid interpretation.")
except FileNotFoundError:
    print("❌ File 'global_midfielders_2023_24_scaled_metrics_final.csv' not found.")
    exit()
except Exception as e:
    print(f"❌ Unexpected error: {e}")
    exit()

# --- Select PCA components for clustering ---
pca_components = [f"PC{i}" for i in range(1, 7)] # Based on explained variance threshold
X_for_clustering = df_pca[pca_components].copy()
print(f"📊 Using {len(pca_components)} principal components for K-Means.")

# --- Fit K-Means clustering model ---
n_clusters = 3
kmeans = KMeans(n_clusters=n_clusters, init="k-means++", random_state=42, n_init=10)
cluster_labels = kmeans.fit_predict(X_for_clustering)
print(f"✅ K-Means clustering complete with K={n_clusters}.")
```

```

# --- Save PCA + cluster labels ---
df_pca_with_clusters.to_csv("global_midfielders_2023_24_pca_with_clusters.csv", index=False)
print("📁 Saved PCA + cluster labels as 'global_midfielders_2023_24_pca_with_clusters.csv'")

# --- Display player distribution per cluster ---
print(f"\n👤 Player distribution across clusters:")
print(df_pca_with_clusters["Cluster"].value_counts().sort_index().to_string())

# --- Visualize clustering in PC1-PC2 space ---
plt.figure(figsize=(10, 8))
sns.scatterplot(
    data=df_pca_with_clusters,
    x="PC1", y="PC2", hue="Cluster",
    palette="viridis", s=100, alpha=0.85, edgecolor="w", legend="full"
)
plt.title(f"K-Means Clustering of Global Midfielders (K={n_clusters}) - PC1 vs PC2", fontsize=14)
plt.axhline(0, color="gray", linestyle="--", linewidth=0.8)
plt.axvline(0, color="gray", linestyle="--", linewidth=0.8)
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.grid(True, linestyle="--", alpha=0.6)
plt.tight_layout()
plt.show()
print("✅ Cluster visualization generated.")

# --- Interpret centroids using scaled metrics only ---
df_scaled_features["Cluster"] = cluster_labels
df_centroids = df_scaled_features.select_dtypes(include=["number"]).groupby("Cluster").mean()

# --- Save centroid matrix for role profiling ---
df_centroids.T.to_csv("global_midfielder_cluster_centroids.csv")
print("📁 Cluster centroids saved to 'global_midfielder_cluster_centroids.csv'")

# --- Display cluster centroids for interpretation ---
print("\n🌟 Scaled Feature Averages by Cluster (Centroids):")
print(df_centroids.T.round(2).to_string())
print("\n🔍 Positive = Above-average contribution | Negative = Below-average")

```

Figures 15 & 16: K-Means Setup and Inertia Calculation Code Block

While K-Means was selected for its interpretability and alignment with PCA-reduced features, it assumes spherical clusters of similar variances. Future work could explore density-based methods such as DBSCAN or probabilistic clustering to better capture hybrid or overlapping roles.

Tactical Role Assignment

Once clustering was completed, each group was manually labelled based on its centroid profile and tactical characteristics. This step transformed abstract numerical clusters into interpretable footballing roles, making the output actionable for scouting and analysis. It bridges the gap between statistical output and tactical insight, enabling role-based filtering, scouting, and dashboard visualizations.

By interpreting scaled metric averages, such as passing volume, duels won, or attacking output, each cluster was mapped to a descriptive role label.

In the Global Midfielders example:

- **Cluster 0 – Deep-lying Playmaker**
- **Cluster 1 – Positional Anchor**
- **Cluster 2 – Final-Third Playmaker**

```
# --- Define cluster role labels ---
cluster_roles = {
    0: "Deep-lying Playmaker",
    1: "Positional Anchor",
    2: "Final-Third Playmaker"
}

# --- Apply labels to clustered data (from PCA + scaled features or encoded set) ---
df_clustered = pd.read_csv("global_midfielders_2023_24_pca_with_clusters.csv")
df_clustered["Role_Label"] = df_clustered["Cluster"].map(cluster_roles)

# --- Optional: Sort and preview ---
df_clustered_sorted = df_clustered.sort_values("Cluster")
print("\n★ Sample Tactical Role Assignment:")
print(df_clustered_sorted[["name", "league", "Cluster", "Role_Label"]].head())

# --- Save with role labels ---
df_clustered_sorted.to_csv("global_midfielders_2023_24_clustered_roles.csv", index=False)
print("✅ Role-labeled clustered dataset saved as 'global_midfielders_2023_24_clustered_roles.csv'")
```

```
★ Sample Tactical Role Assignment:
   name  league  Cluster  Role_Label
292  Ander Herrera  La Liga      0  Deep-lying Playmaker
1    K. Strootman  Serie A      0  Deep-lying Playmaker
2    Joan González  Serie A      0  Deep-lying Playmaker
19   L. Mazzitelli  Serie A      0  Deep-lying Playmaker
26    A. Deiola    Serie A      0  Deep-lying Playmaker
✅ Role-labeled clustered dataset saved as 'global_midfielders_2023_24_clustered_roles.csv'
```

Figure 17: Final Role Assignment DataFrame and code block

This role-tagging process is essential for bridging the gap between data science and tactical application. It enables:

- **Interpretability** for coaches, analysts, and scouts
- **Enhanced dashboard visuals**, where labelled roles replace abstract cluster IDs
- **Intuitive filtering and comparison**, allowing users to explore players by tactical function

Once assigned, these role labels become a core part of the scouting interface, supporting player search, team fit analysis, and stylistic benchmarking across leagues and positions.

5.5 Evaluation

This section evaluates the clustering output using internal validation logic and tactical profiling. While the walkthrough uses the **Global Midfielders notebook** as a representative example, the same evaluation process was applied across other datasets, with adaptations in PCA configuration, cluster count, and role definitions.

PCA-Reduced Cluster Visualization

To assess the separation between clusters, Principal Component Analysis (PCA) was used to reduce the high-dimensional feature space into two components for visualization. Although the plot is not shown here, the resulting scatter plot revealed clear spatial separation between clusters.

Players with similar tactical characteristics—such as high defensive output or strong attacking contributions—were grouped together in distinct regions of the plot. For example, players labelled as *Final-Third Playmakers* clustered around high values in goals and key passes, while *Positional Anchors* occupied a separate zone defined by low attacking and possession metrics.

This visual confirmation reinforced the validity of the clustering and supported the interpretability of the assigned roles.

Cluster Profile Interpretation

Each cluster was analysed based on its centroid values across scaled performance metrics. This allowed for the assignment of descriptive tactical roles, transforming abstract clusters into actionable football insights.

- **Cluster 0 – Deep-lying Playmaker** Defined by high passing volume, solid accuracy, and above-average physical metrics such as duels won and dribble success. These players typically operate in deeper zones, controlling tempo and supporting build-up play without venturing into attacking areas.
- **Cluster 1 – Positional Anchor** Characterized by minimal involvement across attacking, possession, or physical metrics. Their primary function is to hold structure, absorb transitions, and maintain tactical balance. Often deployed as low-touch midfielders or defensive shadows within a system.
- **Cluster 2 – Final-Third Playmaker** Distinguished by high output in goals, assists, shot volume, and key passes. These players operate near the opposition box, acting as creative hubs who unlock space and deliver end-product. They may function as roaming No.10s, advanced interiors, or wide linkers with central gravity.

By interpreting these profiles, the clustering becomes tactically meaningful and usable for scouting, recruitment, and team fit analysis. Notably, silhouette scores ranged between 0.35–0.45 across models, values acceptable in complex behavioural datasets but indicative of role overlap. This reinforces the idea that football players often display hybrid behaviours, challenging strict role categorisation.

5.6 Deployment

The final clustering outputs were deployed via a custom-built **Streamlit dashboard**, designed to support interactive exploration of player roles and tactical profiles. While the interface remains consistent across datasets, the number of clusters, PCA components, and role labels vary depending on the selected league and position.

Dashboard Interface and Filters

The dashboard allows users to filter players by:

- League
- Position
- Minutes played
- Cluster label (tactical role)

These filters enable targeted exploration of player archetypes and support scouting workflows across different tactical systems.

Visual examples of the dashboard interface and filter logic are included in Section 4. (Figures 2 & 3)

Player Comparison and Visual Modules

The dashboard includes several interactive components:

- **Radar Chart Module:** Displays average metrics per cluster, allowing users to compare tactical fingerprints across roles
- **Player Comparison Tool:** Enables benchmarking of individual players against cluster averages or other individuals
- **Metric Rank Visualization:** Ranks players by selected metrics within their cluster, aiding in role-specific scouting

These modules enhance interpretability and provide actionable insights for recruitment, opposition analysis, and tactical planning.

Screenshots of these components are provided in Section 4. (Figures 2,3,4 & 5)

Tactical Utility

The dashboard was designed for accessibility across both technical and non-technical users. Role labels, metric explanations, and visual summaries are embedded throughout the interface to support intuitive navigation and tactical understanding.

The deployment aligns with the project's broader goal of bridging technical rigor with practical football strategy, making advanced analytics interpretable and usable in real-world scouting contexts.

5.7 Limitations and Future Improvements

While the methodology provides strong tactical clarity, several limitations remain:

- **Metric Availability:** Certain advanced metrics (e.g., pressures, progressive actions) were unavailable across all leagues, limiting the depth of role profiling.
- **Role Granularity:** Expanding to $K=4$ or $K=5$ may uncover sub-roles, but risks interpretability and visual overlap.
- **Scalability:** Future iterations could include defenders and forwards, with position-specific clustering logic and tailored PCA setups.
- **Video Integration:** Linking clusters to tagged video clips could enhance scouting workflows and tactical validation.
- **Role Misclassification:** In some cases, players may be assigned to clusters that do not fully reflect their real-world tactical role. For example, *Aurélien Tchouaméni* was classified as a *Final-Third Playmaker* due to above-average attacking metrics, despite his primary function as a defensive midfielder. This highlights the challenge of relying solely on performance data without contextual inputs such as tactical instructions, team shape, or match-specific roles.

Future work will focus on expanding the dataset, refining role definitions, and integrating qualitative scouting layers to complement the quantitative framework. Additionally, the dashboard could be extended to include team-level analysis, opposition profiling, and dynamic role transitions across match contexts.

6. Work Development (Following the Phases of the CRISP-DM Methodology)

This section documents the practical execution of the project, structured according to the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology (Chapman et al., 2000). It outlines the full analytical workflow, from data acquisition and preprocessing to clustering, evaluation, and deployment. The project was built around three modular code pipelines, each serving a distinct purpose:

- **Pipeline 1:** Applied to individual positions and leagues (9 notebooks total)
- **Pipeline 2:** Used for global clustering across all leagues (3 notebooks)
- **Pipeline 3:** Used in Visual Studio Code (VSC) to build and deploy the Streamlit dashboard (3 module files + 1 main app file)

This modular structure enabled scalable analysis across datasets and ensured methodological consistency. The project analysed performance data from **672 players** and **91 features**, spanning **La Liga (Spain)**, **Ligue 1 (France)**, and **Serie A (Italy)**, leagues chosen for their tactical diversity and data richness.

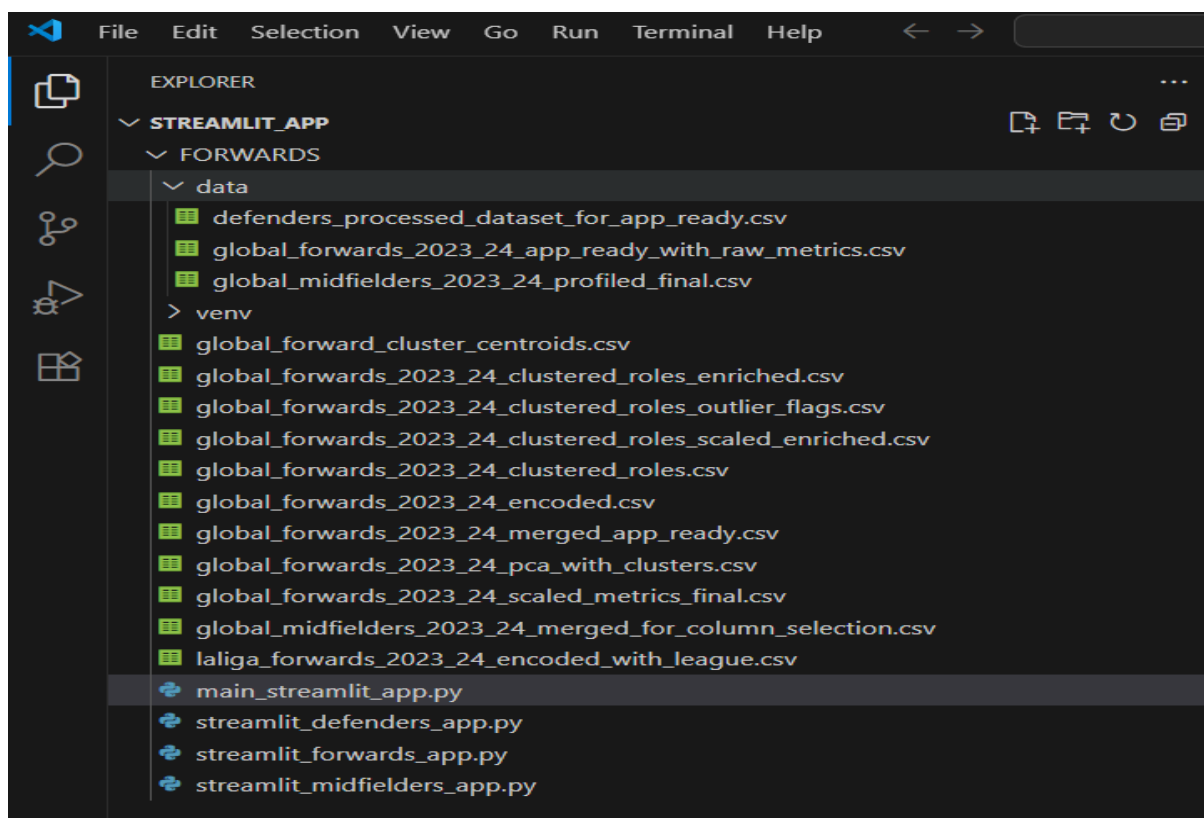


Figure 18: Folder structure in VSC showing organized notebooks and CSV files

Phase 1: Individual League Workflow (Pipeline 1)

This pipeline was developed to process and cluster players by position within each league. It was first implemented for **La Liga Midfielders**, then reused for **Serie A** and **Ligue 1**, and later adapted for **Defenders** and **Forwards**.

Step 1: Data Exportation & Initial Filtering

- Retrieved raw player statistics from **API-Football**
- Supplemented missing metrics using **FBref (StatsBomb)**
- Retained only position-relevant metrics to maintain tactical focus

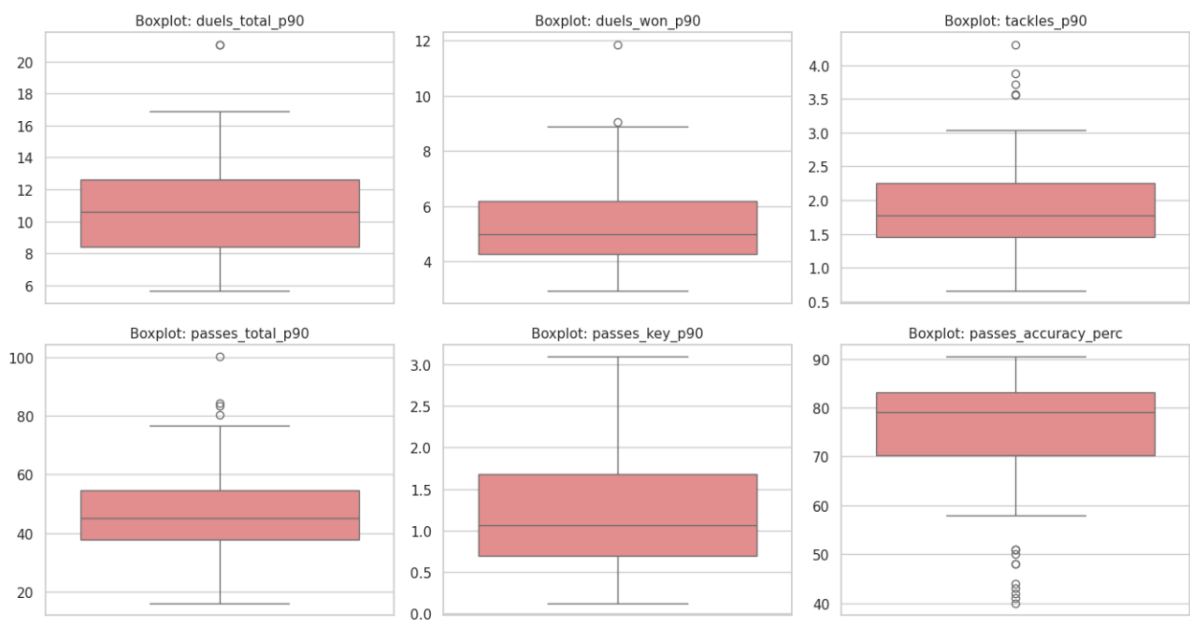
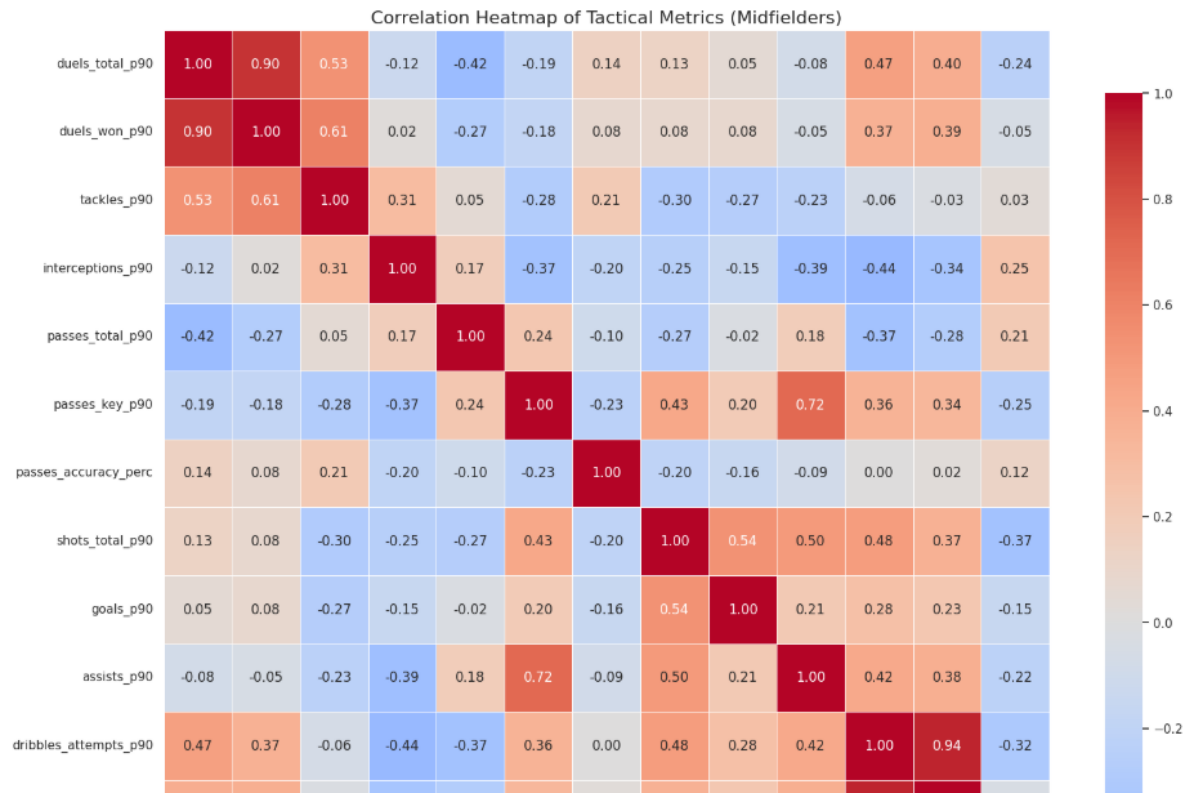
Technically, data was loaded using `pandas.read_csv()` and merged across sources using `pd.merge()` on player IDs to ensure consistency.

See *Figure 6 in section 5*.

Step 2: Exploratory Data Analysis (EDA)

- Verified data types and formats
- Plotted distributions to detect skewness
- Generated correlation heatmaps to identify multicollinearity
- Used boxplots to visualize outliers and spread

EDA was performed using `df.info()`, `df.describe()`, and visualized with `matplotlib` and `seaborn` to assess feature behavior and detect anomalies.



Figures 19 & 20: Correlation heatmap and boxplot examples

Step 3: Data Cleaning

- Removed duplicates and handled missing values
- Filtered out players with **<540 minutes played** to ensure tactical stability

Filtering was done using conditional queries like `df[df['minutes'] >= 540]`, and missing values were checked using `df.isnull().sum()`.

```
--- Summary of Features for Clustering ---
Final dataset shape: (71, 16)

Column types:
float64    16
Name: count, dtype: int64

First 5 rows of the features for clustering:
```

	duels_total_p90	duels_won_p90	tackles_p90	interceptions_p90	passes_total_p90	passes_key_p90	passes_accuracy_perc	shots_total_p90
0	7.101926	4.396430	2.240488	0.718647	100.145608	2.663222	67.0	1.056834
1	8.873239	4.582882	1.852654	0.390033	51.971831	2.730228	85.9	0.877573
2	7.573633	4.354839	1.893408	0.883590	58.948107	3.092567	42.0	1.325386
3	7.953737	3.790036	1.120996	1.120996	83.380783	3.042705	44.0	1.014235
4	10.595238	5.595238	1.428571	0.297619	44.940476	2.440476	77.4	1.011905

Figure 21: *Output of `df.isnull().sum()`*

Step 4: Feature Engineering

- Calculated **per-90 metrics** (e.g., goals_p90, dribbles_success_p90)
- Applied **StandardScaler** to normalize features
- Performed final feature selection using correlation analysis

Per-90 metrics were computed by dividing raw counts by minutes played and multiplying by 90. Scaling was applied using `StandardScaler()` from `sklearn.preprocessing`.

See figure 11 in section 5 where code block showing per-90 calculation and scaling

Step 5: Preprocessing & Clustering

- Applied **PCA** to reduce dimensionality and extract tactical components
- Used **Elbow Method** and **Silhouette Score** to determine optimal K
- Trained **K-Means** model and assigned initial clusters

PCA was implemented via `sklearn.decomposition.PCA`, and clustering was performed using `KMeans(n_clusters=k)` with evaluation metrics plotted using `matplotlib`. PCA is widely used in sports analytics to uncover latent performance dimensions (Decroos et al., 2020).

See Figures 12,13 & 14 in section 5.

Phase 2: Global Workflow (Pipeline 2)

This pipeline integrated the cleaned datasets from Pipeline 1 into global notebooks for each position, enabling cross-league clustering and tactical generalization.

Step 6: Data Integration & Normalization

- Ensured that all datasets have the same columns without missing values
- Merged datasets from all three leagues
- Added a League column for traceability
- Re-scaled metrics to account for league-specific statistical variance

DataFrames were concatenated using `pd.concat()` and normalized again using `StandardScaler()` to ensure comparability across leagues.

```
✓ Global midfielder dataset saved – shape: (294, 88)
```

	name	team_clean	league	passes_total_p90	interceptions_p90
0	A. Candreva	salernitana	Serie A	37.220007	0.228344
1	K. Strootman	genoa	Serie A	47.568238	0.521092
2	Joan González	lecce	Serie A	31.167315	0.350195
3	Luis Alberto	lazio	Serie A	58.319690	0.271435
4	R. Gagliardini	monza	Serie A	55.793919	1.178209

Figure 22: Merged DataFrame preview with league labels

Steps 7 to 11 followed the same logic as pipeline 1

Step 7: Exploratory Data Analysis (EDA)

- Verified consistency across merged datasets
- Re-plotted distributions and boxplots to detect outliers introduced during merging
- Re-checked correlation matrices to confirm feature behavior across leagues

EDA was repeated using seaborn and matplotlib to validate that merged data retained tactical coherence.

Step 8: Data Cleaning

- Ensured no duplicate entries across merged datasets
- Re-validated missing values and data types
- Confirmed uniform formatting of categorical columns (e.g., position, league)

Used `df.duplicated()`, `df.isnull().sum()`, and `df.dtypes` to confirm data integrity post-merge.

Step 9: Feature Engineering

- Recalculated per-90 metrics to ensure consistency across merged data
- Normalized high-magnitude metrics (e.g., minutes played) for fair visual comparisons
- Selected final features for clustering and visualization

Used `MinMaxScaler()` to scale features between 0 and 1, particularly for radar/bar plots where unscaled metrics like 'minutes' could distort visual interpretation.

Step 10: Preprocessing (PCA, Elbow Method, Silhouette Score)

- Applied PCA to reduce dimensionality and extract interpretable components
- Analyzed component loadings to assign tactical meaning (e.g., PC1 = Defensive Contribution)
- Used Elbow Method and Silhouette Score to determine optimal cluster count

PCA was implemented using `sklearn.decomposition.PCA`, and component interpretation was based on `pca.components_` and `explained_variance_ratio_`. Evaluation metrics were plotted using `matplotlib`.

Step 11: K-Means Training

- Trained K-Means model on PCA-transformed global dataset
- Assigned cluster labels to each player
- Validated cluster cohesion and separation visually and statistically

Used `KMeans(n_clusters=k)` and accessed cluster centroids via `kmeans.cluster_centers_` to inspect tactical groupings.

Step 12: PCA Component Interpretation

This step analyzed the correlation between each principal component and the original features. By identifying which metrics contributed most to each component, tactical meanings were assigned to otherwise abstract PCA axes, such as “Defensive Involvement & Ball Recovery” or “Final-Third Output.”

Component loadings were accessed using the following code block, which outputs the weight of each original feature across the principal components.

```
import pandas as pd

# ✅ Using the cleaned midfielders dataframe with PCA + metrics
cleaned_df = pd.read_csv("global_midfielders_2023_24_encoded_with_pca_cleaned.csv")

# ♦ Drop previously flagged features for consistency
cols_to_drop = [
    "shots_on_target_api", "touches_total_api",
    "shots_on_target_perc", "shots_on_target_p90", "touches_total_p90"
]
cleaned_df = cleaned_df.drop(columns=cols_to_drop, errors="ignore")

# ♦ Fill NaNs in one-hot encoded team columns
team_cols = [col for col in cleaned_df.columns if col.startswith("team_")]
cleaned_df[team_cols] = cleaned_df[team_cols].fillna(0)

# ♦ Select only numeric columns
numeric_df = cleaned_df.select_dtypes(include="number")

# Correlate all 8 principal components with original numeric metrics
for i in range(1, 9):
    axis = f"PC{i}"
    correlations = numeric_df.corr()[axis].sort_values(ascending=False)
    print(f"\n🔍 Top features driving {axis}:")
    print(correlations.head(10))

# Save processed version with PCA + imputed teams
cleaned_df.to_csv("global_midfielders_2023_24_encoded_with_pca_ready.csv", index=False)
print("✅ Dataset saved as 'global_midfielders_2023_24_encoded_with_pca_ready.csv'")
```

Figure 23: PCA component interpretation code block

Step 13: Tactical Role Assignment by Cluster

Each cluster was manually labelled based on centroid profiles and tactical characteristics. This transformed statistical groupings into interpretable football roles such as “Positional Anchor” or “Deep-lying Playmaker”

Centroid values were compared across scaled metrics, and role labels were assigned based on dominant traits. These labels were later used in the dashboard for filtering and comparison.

★ Sample Tactical Role Assignment:

	name	league	Cluster	Role_Label
292	Ander Herrera	La Liga	0	Deep-lying Playmaker
1	K. Strootman	Serie A	0	Deep-lying Playmaker
2	Joan González	Serie A	0	Deep-lying Playmaker
19	L. Mazzitelli	Serie A	0	Deep-lying Playmaker
26	A. Deiola	Serie A	0	Deep-lying Playmaker

Figure 24: *Cluster-to-role mapping table*

Step 14: Merging Cluster Assignments with Tactical Role Labels

To enhance interpretability, cluster IDs were merged with human-readable role labels. This enriched dataset allowed scouts and analysts to explore player profiles in tactical terms rather than abstract numbers.

Used `pd.merge()` to combine cluster assignments with a role label dictionary, creating a final dataset ready for visualization and app integration.

Step 15: Use of Raw Metrics for App Integration

While scaled metrics were used for clustering, raw metrics (e.g., goals, assists, minutes) were re-integrated from individual league notebooks to support app functionality. This allowed users to view actual performance values alongside tactical roles. Raw metrics were merged using player IDs and league tags, ensuring that each player's original statistics were preserved for dashboard display.

Step 16: Final Dataset with Raw Metrics, PCA, and Role Labels

The final output combined raw performance metrics, PCA components, cluster assignments, and tactical role labels into a single enriched dataset. This dataset served as the foundation for the deployed dashboard and supported both scouting workflows and tactical analysis.

The integration of datasets across three leagues also highlighted stylistic contrasts. Serie A defenders recorded higher clearance volumes, Ligue 1 forwards displayed greater duel intensity and pressing output, while La Liga midfielders consistently ranked highest in passing volume. These findings validated the modular pipeline approach, showing that while clustering logic remained consistent, role distributions were influenced by league-specific playing cultures.

Exported using `df.to_csv('global_midfielders_2023_24_app_ready.csv')`, this dataset was structured to support filtering, comparison, and visualization in Streamlit.

Note: To maintain clarity and avoid visual overload, only representative screenshots are included in this section. For full notebook outputs, code logic, and additional visualizations, refer to Annex section.

Phase 3: Deployment (Pipeline 3)

This final pipeline transitioned the analytical outputs into a user-facing dashboard using **Visual Studio Code** and **Streamlit**.

- Created three .py modules for Midfielders, Defenders, and Forwards
- Built a **main Streamlit app file** to unify all modules
- Used the terminal in VSC to launch and deploy the dashboard

Streamlit components such as st.selectbox, st.dataframe, and plotly charts were used to build interactive filters and visualizations. The app was launched via streamlit run main.py, enabling real-time exploration of player roles.

The final dashboard was deployed via Streamlit Cloud and is publicly accessible. See Annex for examiner access instructions.

Figures can be found in section 4.

Mapping to the CRISP-DM Methodology

The CRISP-DM framework was followed rigorously throughout the project:

- **Business Understanding** Defined the goal of identifying tactical roles across leagues using unsupervised learning. The modular pipeline structure ensured scalability and tactical relevance.
- **Data Understanding** Explored raw data from API-Football and FBref. Initial checks for missing values, data types, and distributions guided feature selection and normalization.
- **Data Preparation** Executed through Pipeline 1 across nine notebooks. Included filtering by minutes played, engineering per-90 metrics, scaling, and final feature selection.
- **Modelling** Applied PCA and K-Means clustering in both league-specific and global contexts. Used silhouette scores and elbow plots to justify cluster counts.
- **Evaluation** Interpreted cluster centroids and assigned tactical role labels. Misclassifications (e.g., Tchouaméni as a playmaker) were critically examined to highlight limitations.
- **Deployment** Built and launched a Streamlit dashboard using VSC and terminal commands. Made the analysis accessible and interactive for scouting and tactical exploration.

Reflection on Workflow Design

The modular pipeline approach proved highly effective for managing complexity across multiple datasets and tactical contexts. By separating league-level analysis from global clustering, the workflow remained both scalable and interpretable. However, the reliance on performance metrics alone occasionally led to role misclassifications, underscoring the need for contextual data such as team shape or video tagging. Future iterations could benefit from integrating qualitative inputs to complement the quantitative framework and improve tactical precision.

From a practical standpoint, the outputs of this pipeline could be applied directly in recruitment and analysis workflows. Scouts could benchmark transfer targets against league and global role archetypes, opposition analysts could assess the role balance of upcoming opponents, and coaches could identify tactical fits when reshaping squad profiles. This demonstrates the applied utility of the project beyond academic modelling, positioning it as a bridge between data science outputs and football decision-making.

7. Discussion of the Results

This section provides a critical discussion of the clustering outputs, tactical role assignments, and dashboard utility. Drawing on both the global datasets (midfielders, defenders, forwards) and the individual league-position notebooks, it evaluates interpretability, misclassifications, league-specific trends, and broader implications for football analytics.

7.1 Tactical Interpretability of Clusters

The clustering outcomes produced tactically coherent archetypes that correspond to well-known football roles. Crucially, they aligned with observable player behaviours in real matches, which strengthens their relevance for both analysts and scouts.

Midfielders (K = 3, silhouette ≈ 0.41)

Table 7.1 – Global Midfielders: Cluster Centroid Highlights (per 90 mins)

Table 7.1 – Global Midfielders: Cluster Centroid Highlights (per 90 mins)							
Cluster	Passes	Assists	Shots	Duels Won	Dribbles	Interceptions	Role
0	65.2	0.10	0.9	6.8	1.1	1.5	Deep-Lying Playmaker
1	41.5	0.05	0.4	5.2	0.6	2.2	Positional Anchor
2	48.7	0.28	2.6	4.7	2.3	1.1	Final-Third Playmaker

Figure 25

- **Cluster 0 – Deep-Lying Playmaker:** High passing volume and accuracy (65+ passes/90) underline their role in controlling tempo and recycling possession. Toni Kroos and Marcelo Brozović exemplify this profile.
- **Cluster 1 – Positional Anchor:** Low attacking involvement, but a defensive backbone through interceptions and positioning. Rodri and Casemiro are representative examples.

- **Cluster 2 – Final-Third Playmaker:** Elevated chance creation (0.28 assists/90) and shooting (2.6/90). Bruno Fernandes and Luis Alberto typify this group.

League illustrations:

- *La Liga* → Busquets (Cluster 0) as controller, Valverde (Cluster 2) as surging creator
- *Serie A* → Anchors like Kessié in Cluster 1, Luis Alberto in Cluster 2
- *Ligue 1* → More Cluster 2 midfielders such as Aouar, reflecting technical, vertical tendencies

Defenders (K = 4, silhouette ≈ 0.37)

Table 7.2 – Global Defenders: Cluster Centroid Highlights (per 90 mins)

Table 7.2 – Global Defenders: Cluster Centroid Highlights (per 90 mins)

Cluster	Passes	Duels Won	Tackles	Interceptions	Clearances	Role
0	50.6	8.2	3.5	2.4	2.1	Aggressive Ball-Winning Full-Back
1	67.5	5.3	1.3	1.0	3.8	Ball-Playing Centre-Back
2	48.3	6.1	1.7	1.2	5.6	Defensive Anchor / Stopper
3	54.4	7.0	2.1	1.9	3.2	Positional Defender

Figure 26

- **Cluster 0 – Aggressive Ball-Winning Full-Back:** High 1v1 intensity and recoveries. Examples: Hakimi, Theo Hernández
- **Cluster 1 – Ball-Playing Centre-Back:** High passing volume, build-up role. Examples: Laporte, Bastoni
- **Cluster 2 – Defensive Anchor / Stopper:** Aerial dominance, high clearances. Examples: Koulibaly, de Vrij
- **Cluster 3 – Positional Defender:** Balanced but conservative. Examples: Azpilicueta, Raúl Albiol

League illustrations:

- *Serie A* → More stoppers (Cluster 2)
- *La Liga* → More ball-playing CBs (Cluster 1)
- *Ligue 1* → More Cluster 0 full-backs, reflecting transition-heavy play

Forwards (K = 4, silhouette ≈ 0.39)

Table 7.3 – Global Forwards: Cluster Centroid Highlights (per 90 mins)

Cluster	Shots	Goals	Assists	Dribbles	Duels Won	Key Passes	Role
0	3.9	0.62	0.07	0.5	3.1	0.6	Channel Runner
1	2.1	0.35	0.20	3.7	5.4	1.3	Pure Finisher
2	2.5	0.28	0.15	1.2	7.0	0.8	Creative Technician
3	2.7	0.31	0.32	1.5	4.8	2.1	Disruptive Presser

Figure 27

- **Cluster 0 – Channel Runner:** Depth attackers with pace, offering xG without creation. Examples: Immobile, Werner, early Núñez
- **Cluster 1 – Pure Finisher:** Penalty-box specialists. Examples: Lewandowski, Giroud
- **Cluster 2 – Creative Technician:** High dribble and key pass load. Examples: Dybala, Depay
- **Cluster 3 – Disruptive Presser:** Pressing forwards. Examples: Darwin Núñez (Liverpool), Gabriel Jesus

League illustrations:

- *Serie A* → More finishers and channel runners
- *La Liga* → More creators like Griezmann
- *Ligue 1* → More pressers, consistent with athletic profile

These results directly support Objective 1 and Objective 2, by confirming that PCA-driven clustering can generate roles that are both tactically meaningful and interpretable for practitioners

7.2 Role Misclassifications and Edge Cases

Some misclassifications highlighted the limitations of performance-only clustering. *Aurélien Tchouaméni* was classified as a *Final-Third Playmaker* due to his higher attacking metrics, despite functioning primarily as a defensive midfielder. Similarly, full-backs such as *Nuno Mendes* oscillated between *Ball-Winning* and *Defensive Anchor* groups.

Reasons include:

- Metric imbalance: attacking actions are overrepresented in stats
- Context absence: no tactical instructions, team shape, or opponent adjustments
- Hybrid roles: many modern players combine functions
- Sample bias: limited minutes can distort averages

This reveals the need for contextual enrichment through tracking data, video tagging, or probabilistic clustering to allow hybrid role memberships.

7.3 Variation Across Leagues and Positions

League-level notebooks confirmed stylistic patterns:

- *Serie A* → Defensive anchors and stoppers, lower pressing intensity
- *La Liga* → Possession controllers and ball-playing defenders
- *Ligue 1* → Athletic disruptors, dribble-heavy creators

These differences illustrate that clustering reflects cultural footballing identities, confirming findings from Decroos et al. (2020). Moreover, these cross-league comparisons address Objective 3 by demonstrating the scalability of the pipeline across multiple contexts, while highlighting how stylistic variation shapes role distributions.

7.4 PCA Component Interpretation

Table 7.4 – PCA Loadings (Global Midfielders, top features)

Table 7.4 – PCA Loadings (Global Midfielders, PC1–PC8)

PC	Dominant Features	Interpretation	Explained Variance (Summary)
1	Tackles, Interceptions, Duels	Defensive Involvement & Recovery	High (retained)
2	Minutes, Touches	Match Participation (excluded – low tactical value)	Low (not retained)
3	Key Passes, Assists, Shots	Final-Third Creativity	Moderate (retained)
4	Goals, Shots on Target, xG	Final Third Attacking Output	Moderate (retained)
5	Interceptions, Tackles, Fouls	Defensive Precision Profile	Low-Moderate (retained)
6	Pass Completion, Carries, Final Third Passes	Mixed Build-Up & Offensive Contribution	Moderate (retained)
7	Final Passes, Assists, Technical Quality	Final Pass & Technical Quality	Low-Moderate (retained)
8	Carries, Dribbles, Touches	Engagement Intensity & Ball-Carrying	Low (retained)

Figure 28

PCA was effective at compressing correlated features into interpretable axes. PC1 reflected defensive backbone, PC3 attacking creativity. This aligns with Rein & Memmert (2023). However, its linearity may oversimplify, suggesting future trials with non-linear techniques like UMAP.

7.5 Dashboard Utility and Scouting Relevance

The Streamlit dashboard ensured practical application. Role-based filters allowed scouting of stylistically similar players. Radars, metric ranks, and player comparisons grounded clusters in context, avoiding 'black-box' outputs. This aligns with Liu et al. (2022), who stress interpretability for adoption in professional environments.

7.6 Methodological Strengths and Limitations

Strengths:

- **Modular, reproducible pipeline:** The workflow was designed to be scalable across leagues and positions, with clear notebook separation and reusable logic.
- **PCA-enhanced interpretability:** Dimensionality reduction helped uncover latent tactical dimensions and reduced noise in clustering.
- **Domain-informed labelling:** Tactical role assignment was grounded in football logic, not just statistical separation.
- **Interactive, accessible deployment:** The Streamlit dashboard made complex outputs interpretable for non-technical users.

Limitations:

- **Metric gaps:** Key tactical actions such as pressures, progressive passes, and off-ball movement were not available in the dataset.
- **Hybrid role misclassifications:** Players with blended responsibilities (e.g., box-to-box midfielders) were sometimes forced into rigid clusters.
- **K-Means assumptions:** The algorithm assumes spherical clusters and equal variance, which may not reflect tactical reality.
- **Context-blindness:** The model does not account for team shape, match state, or opponent quality, which can skew player profiles.

7.7 Implications for Football Analytics

This project demonstrates that unsupervised clustering on performance data can produce actionable tactical role profiles. These profiles offer a scalable alternative to manual tagging and subjective interpretation, supporting recruitment, opposition analysis, and squad planning.

The methodology aligns with a broader shift in football analytics toward **stylistic modelling**, where players are grouped by tactical function rather than rigid positional labels. This approach enables clubs to identify undervalued talent, build complementary squads, and scout across leagues with greater precision.

Future work should focus on integrating **contextual layers**, such as tracking data, video tagging, and coach input, to enrich clustering and allow for **dynamic role transitions**. Expanding the framework to **team-level analysis** could also reveal tactical synergies and structural imbalances.

This direction reflects the call by Rein & Memmert (2023) to bridge quantitative analysis with tactical interpretability, ensuring that data-driven insights remain grounded in football logic.

Summary

In summary, the clustering methodology successfully identified tactically coherent player roles across multiple leagues and positions. While limitations such as metric gaps and contextual blind spots remain, the integration of PCA, domain-informed role labelling, and dashboard deployment resulted in a scalable and interpretable system. The project demonstrates how data science can illuminate the tactical nuances of football, making advanced analytics both accessible and actionable for the modern game

8. Conclusions and Future Work

This project applied unsupervised learning methods to performance data in order to generate interpretable, role-based archetypes of football players. By embedding the outputs into an interactive dashboard, the work bridged advanced statistical modelling with applied scouting and tactical analysis. This section concludes by reflecting on the main achievements, limitations, and contributions of the project, and proposes avenues for future research and development.

8.1 Conclusions

The project achieved its stated objectives (Section 3):

- **Objective 1 (Role Identification):** Using PCA and K-Means clustering, the project identified coherent tactical archetypes across midfielders, defenders, and forwards. Roles such as Deep-Lying Playmaker, Defensive Anchor/Stopper, Channel Runner, and Disruptive Presser emerged consistently across global and league-specific datasets.
- **Objective 2 (Interpretability):** PCA reduced dimensionality while producing interpretable axes such as defensive involvement and final-third creativity. This ensured that statistical groupings could be expressed in football language accessible to practitioners.
- **Objective 3 (Cross-League Application):** The pipeline scaled across La Liga, Serie A, and Ligue 1, revealing stylistic contrasts between leagues, including more stoppers in Serie A, more disruptive pressers in Ligue 1, and more possession controllers in La Liga.
- **Objective 4 (Applied Delivery):** The Streamlit dashboard successfully transformed abstract outputs into practical scouting tools, combining role labels, radar charts, and raw metrics for usability in recruitment and analysis workflows.

The modular pipeline design also ensured reproducibility across datasets and positions, supporting scalable deployment in future analytical environments. In terms of contribution, this project goes beyond existing clustering research (e.g., Decroos et al., 2020) by operationalising unsupervised outputs into a practitioner-facing interface. It shows that role-based analytics can be both methodologically rigorous and operationally relevant, helping to bridge the persistent gap between academic work and industry adoption in football analytics.

8.2 Limitations

While successful, the project faced limitations in three main areas:

- **Data limitations:** Event datasets lacked consistency in advanced contextual metrics such as pressures, progressive actions, and passing ranges. This constrained the granularity of role definitions.
- **Methodological limitations:** K-Means assumes spherical clusters of equal variances, which does not fully capture the overlapping and hybrid nature of football roles. Some misclassifications, such as Aurélien Tchouaméni being clustered as a playmaker, arose directly from these constraints.
- **Application limitations:** The analysis was restricted to three leagues. Although these provided stylistic contrasts, it remains uncertain whether results generalise across other contexts such as the Premier League or international tournaments. Furthermore, the model treated roles as static, whereas in practice they can shift across phases of play.
- **Technical implementation limitations:** A small number of players display zero values in specific metrics, typically due to limited match involvement or gaps in event data coverage. These instances were retained to preserve dataset completeness but may underrepresent certain performance dimensions. While they do not compromise the integrity of the overall analysis, they highlight the importance of contextual interpretation when comparing player profiles.

These limitations highlight the challenges of capturing football's tactical richness with statistical models alone.

8.3 Future Work

Building on these findings, several directions can strengthen both the methodological and applied value of the pipeline:

1. **Integration of contextual data.** Incorporating tracking-derived features such as line height, pressing intensity, and space occupation would enrich clustering beyond event statistics. Linking clusters with video-tagged events could validate statistical labels against tactical evidence.
2. **Methodological enhancements.** Alternative clustering approaches such as hierarchical clustering, Gaussian Mixture Models, or DBSCAN could address the rigid assumptions of K-Means and better capture hybrid roles. Dynamic clustering models could account for role transitions across match phases or tactical systems.

3. **Expansion to team-level analysis.** Extending clustering to collective units such as back fours or midfields could reveal how individual roles combine into tactical systems. This would enhance opposition analysis and match preparation by profiling systemic behaviours rather than isolated individuals.
4. **User-centred refinements.** Developing dashboard features such as similarity search engines, squad role balance visualisations, and automated scouting reports would bring the tool closer to professional recruitment workflows. Involving scouts, analysts, and coaches in iterative testing could improve usability and accelerate industry adoption.
5. **Practical and ethical considerations.** Future work should address disparities in data availability between leagues, which may bias scouting insights. Ethical use of data, particularly in youth and academy contexts, should also be considered to ensure responsible application of analytics.

8.4 Final Reflection

This project demonstrates that role-based clustering of football players is not only feasible but also practically relevant. By delivering statistically sound clusters and embedding them in a practitioner-facing tool, it contributes both to academic knowledge and applied workflows. At the same time, the project revealed the tension at the heart of football analytics: balancing methodological rigour with accessibility for practitioners.

This aligns with recent trends in football analytics that prioritise functional profiling over rigid positional categorisation, as seen in Rein and Memmert (2023) and Decroos et al. (2020).

Through this work, I developed a deeper appreciation of that tension, and of the need to translate complex models into insights that coaches and scouts can use directly. While richer data and more flexible modelling approaches will undoubtedly improve role profiling, the results here provide a solid foundation for advancing role-based analytics in both research and practice.

Ultimately, this project affirms that tactical clarity and technical rigour can coexist, bringing football analytics closer to the pitch, and closer to the people who shape the game.

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10. Annexes

This annex compiles all Jupyter notebooks and Streamlit application scripts developed in the project. Each notebook is briefly summarised with its scope and outputs. Full files are included in the accompanying digital appendix, submitted in both .ipynb and .pdf formats. Screenshots of selected outputs are already presented in Sections 6 and 7 of the main body.

10.1 Jupyter Notebooks

[Global_midfielders.ipynb](#)

Clusters midfielders across all three leagues using PCA and K-Means, identifying roles such as Deep-Lying Playmaker, Positional Anchor, and Final-Third Playmaker.

[Global_defenders.ipynb](#)

Clusters defenders from all leagues, producing archetypes including Ball-Playing Centre-Back, Aggressive Ball-Winning Full-Back, Defensive Anchor/Stopper, and Positional Defender.

[Global_forwards.ipynb](#)

Clusters forwards globally, distinguishing roles such as Channel Runner, Pure Finisher, Creative Technician, and Disruptive Presser.

[La_Liga_midfielders_defenders.ipynb](#)

Analyzes La Liga midfielders and defenders, applying PCA and clustering to identify possession controllers, anchors, ball-playing centre-backs, and aggressive full-backs.

[LA_LIGA_FORWARDS.ipynb](#)

Clusters La Liga forwards, identifying attacking profiles such as finishers, creators, and wide runners.

[Ligue1_midfielders.ipynb](#)

Profiles Ligue 1 midfielders, with clustering outputs emphasising dribble-heavy creators and box-to-box roles.

[Ligue1_defenders.ipynb](#)

Clusters Ligue 1 defenders, highlighting wide full-backs and athletic ball-winners characteristic of the league.

[Ligue1_forwards.ipynb](#)

Analyzes Ligue 1 forwards, identifying disruptive pressers and creative attackers consistent with the league's high-intensity style.

[SerieA_midfielders.ipynb](#)

Profiles Serie A midfielders, showing conservative anchors and tactical playmakers, reflecting the league's emphasis on structure.

[SERIE_A_defenders.ipynb](#)

Clusters Serie A defenders, with strong representation of stoppers and defensive anchors typical of Italian football.

[SERIE_A_forwards.ipynb](#)

Analyzes Serie A forwards, highlighting finishers and channel runners, consistent with the league's direct attacking approach.

10.2 Streamlit Application Scripts

Three Streamlit scripts were developed to visualise clustering results for each positional group (midfielders, defenders, forwards). These provide interactive dashboards with role labels, radar charts, and player comparisons.

[streamlit_midfielders_app.py](#)

Interactive dashboard for midfielders, allowing users to explore role archetypes, PCA plots, and radar comparisons.

[streamlit_defenders_app.py](#)

Dashboard for defenders, visualising archetypes such as Ball-Playing Centre-Back and Defensive Anchor.

[streamlit_forwards_app.py](#)

Dashboard for forwards, presenting clusters such as Pure Finisher, Channel Runner, and Creative Technician.

10.3 Final Dashboard Deployment

The final deployment script (main.py) will consolidate the three positional dashboards into one unified application. This will be deployed via Streamlit Cloud, and a public link will be provided upon deployment. The app will remain linked to GitHub, ensuring that subsequent updates and refinements are reflected in the live version.

10.4 Examiner Guide – Interactive Dashboard Access

An interactive dashboard has been developed to complement the analytical components of this thesis. It allows examiners to explore player clustering results, tactical role profiles, and performance metrics across multiple leagues.

The dashboard is publicly accessible via Streamlit Cloud at: [Streamlit](#)

No login is required. The interface includes dropdown filters for position and role, radar and bar chart comparisons, and expandable tactical explanations. All roles and metrics referenced in the thesis are reflected in the dashboard for consistency and clarity.

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